

USER MANUAL

Data Weaver – Distributed Agentic Data Pipeline

Version Information

Field	Details
Product Name	Data Weaver
Project Type	Distributed Agentic Data Pipeline
Version	Sprint 3 Release
Platform	Web-Based Dashboard
Developed By	Team 2
UI Framework	Streamlit
Backend	FastAPI + LangGraph

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1. Introduction

Data Weaver is an AI-powered distributed data integration platform designed to automate:

- Data cleaning
- Schema mapping
- Dataset merging
- Relationship detection
- Validation enforcement
- Data documentation generation

The platform uses a **two-agent architecture**:

Agent	Responsibility
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Architect Agent	Infers schema meaning and relationships
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Engineer Agent	Executes transformations and validations
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The system is designed to solve enterprise “data swamp” problems caused by fragmented and inconsistent datasets.

2. System Overview

The platform enables users to:

- Upload multiple CSV datasets
- Automatically detect PK/FK relationships
- Generate ER diagrams
- Clean and merge datasets
- Validate transformations using invariant checks
- Download clean datasets and data dictionaries

Core modules include:

- Ingestion & Scout
- Architect Agent

- Engineer Agent
 - Retry & Repair Loop
 - Invariant Enforcement
 - Loader & Storage
 - Dashboard Visualization
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3. Key Features

3.1 Intelligent Schema Detection

- Detects relationships between datasets
- Identifies primary and foreign keys

Example:

cust_id ↔ customer_id ↔ user_ref

3.2 Automated ERD Generation

Creates interactive Entity Relationship Diagrams automatically.

3.3 Data Cleaning

Supports:

- Duplicate removal
 - Schema casting
 - Null handling
 - Data normalization
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3.4 Deterministic Validation

Ensures reliability using:

- Row count validation
 - Schema validation
 - Null threshold monitoring
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3.5 Real-Time Monitoring

Displays:

- Pipeline execution progress
 - Validation logs
 - Success/failure indicators
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3.6 Downloadable Outputs

Users can export:

- Clean datasets
 - Validation logs
 - Data dictionaries
-

4. User Roles & Personas

The platform supports different enterprise users.

Persona	Primary Goal
Data Engineers	Automate ETL and schema mapping
BI Analysts	Generate clean analytics-ready datasets
CTO / Data Leadership	Ensure governance and scalable AI infrastructure

5. Getting Started

5.1 Accessing the Application

Open the dashboard in a browser:

<http://localhost:8501>

5.2 Supported File Types

Format Supported

CSV	Yes
Excel	No
JSON	Future Release

6. Dashboard Overview

The dashboard contains three primary tabs:

Tab	Purpose
Modeling	ERD and schema understanding
Pipeline	Processing and validation tracking
Output	Download processed files

7. Uploading Datasets

Step 1: Navigate to Upload Section

Locate the upload panel on the dashboard.

Step 2: Upload CSV Files

Users may upload:

- Single dataset
 - Multiple related datasets
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Step 3: Confirm Upload

The system validates:

- File format
- Missing headers
- Invalid records

Sprint functionality includes:

- CSV upload validation
 - Clear error reporting
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8. Data Validation

After upload, the platform performs validation checks.

Validation Types

Validation	Description
Missing Fields	Detects null or missing columns
Invalid Data Types	Verifies schema consistency
Structural Validation	Ensures readable dataset structure

Validation Results

Indicator Meaning

-  Pass Validation successful
 -  Fail Validation failed
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9. Schema Mapping & ERD Generation

The Architect Agent analyzes datasets and identifies:

- Table relationships
- Primary keys
- Foreign keys
- Matching columns

Features implemented in Sprint 1 & 2 include:

- PK/FK detection
 - Schema mapping
 - Interactive ERD visualization
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Viewing the ERD

Navigate to:

Modeling Tab → ER Diagram Viewer

The ERD visually displays:

- Tables
- Relationships
- Keys
- Data connections

10. Data Transformation Pipeline

The Engineer Agent executes:

- Joins
- Deduplication
- Schema casting
- Cleaning operations

Supported transformations include:

- Merge operations
 - Duplicate removal
 - Type conversion
 - Pipeline retries on failure
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Running the Pipeline

Click:

Run Pipeline

The system automatically:

1. Cleans data
 2. Merges datasets
 3. Validates transformations
 4. Stores outputs
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11. Validation & Invariant Checks

The system enforces deterministic validation rules.

11.1 Row Count Validation

Ensures joins do not create invalid record explosions.

Concept used:

$$R_{\text{final}} \leq (R_A \times R_B)$$

11.2 Schema Validation

Verifies output schema matches expected schema types.

11.3 Null Threshold Detection

Flags unusual spikes in missing values.

12. Pipeline Monitoring

The Pipeline Dashboard displays:

- Real-time execution status
- Validation logs
- Processing timeline
- Error tracking

Sprint features implemented include:

- Real-time monitoring
 - Failure traceability
 - Logging visibility
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13. Downloading Outputs

After successful execution, users can download:

Output	Description
Clean Dataset	Final processed CSV
Data Dictionary	Metadata documentation
Validation Logs	Pipeline execution logs

Download Location

Navigate to:

Output Tab → Downloads

14. Data Dictionary

The system auto-generates metadata documentation containing:

- Column names
- Data types
- Table descriptions
- Relationship mappings

This improves:

- Governance
- Transparency
- Analytics readiness

15. Error Handling

Common Errors

Error	Cause	Solution
Upload Failed	Invalid CSV	Re-upload proper CSV
Join Failure	Missing key columns	Validate schema
Validation Failure	Null/type mismatch	Clean source data
ERD Missing	Relationship not inferred	Standardize column names

16. Best Practices

Recommended usage guidelines:

- Use meaningful column names
- Ensure consistent IDs across files
- Avoid duplicate columns
- Validate schema before pipeline execution
- Upload clean CSV files whenever possible

17. Troubleshooting

Issue: Pipeline Stops Unexpectedly

Possible causes:

- Invalid schema
- Null threshold exceeded

- Unsupported data types

Solution

- Review pipeline logs
 - Correct source dataset
 - Re-run pipeline
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Issue: Incorrect Relationship Detection

Solution

Use standardized naming conventions:

customer_id

order_id

product_id

18. Frequently Asked Questions

Q1. Can I upload multiple datasets?

Yes. The platform supports multiple CSV uploads.

Q2. Does the system support Excel files?

Currently, only CSV files are supported.

Q3. Can the system automatically detect relationships?

Yes. The Architect Agent performs automatic schema inference.

Q4. What happens if validation fails?

The pipeline stops execution and displays detailed logs.

Q5. Can I download processed outputs?

Yes. Clean datasets and reports are downloadable.

19. Conclusion

Data Weaver simplifies enterprise data integration by combining:

- AI-driven schema understanding
- Deterministic validation
- Distributed processing
- Interactive visualization
- Automated documentation

The platform enables organizations to build reliable, scalable, and transparent data pipelines while minimizing manual ETL effort.